

Research Plan for Master's Program

Nha Tran Van

September 2026

Title of Research Project

Combining Deep Reinforcement Learning, Monte Carlo Tree Search, and the DCA Algorithm to Enhance Optimization Efficiency and Generalization in Chip Macro Placement

1 Background and Motivation

1.1 Research Background

The explosive development of artificial intelligence in recent years, coupled with the high demand for semiconductor chips, has driven the need to design increasingly large and complex microchips. In the chip design flow, Chip Macro Placement is a critical optimization problem, as the quality of macro placement directly impacts the Half-Perimeter Wirelength (HPWL), Congestion, power consumption, and overall performance of the microchip.

In recent years, Reinforcement Learning (RL) has demonstrated outstanding potential in automating the Chip Macro Placement problem [1]. However, current research still faces three main limitations: (i) very high training costs due to the requirement of a large number of interactions with the environment; (ii) limited generalization capability when applied to chip designs not seen during training; and (iii) significantly reduced effectiveness when the scale or structure of the netlist changes.

To thoroughly resolve these challenges, this research proposes a hybrid framework combining Deep Reinforcement Learning (DRL), Monte Carlo Tree Search (MCTS), and the Difference of Convex Functions Algorithm (DCA). In this framework, DRL plays the role of representation learning and guiding the search; MCTS performs a global tree search to enhance generalization capabilities on novel chip structures; and DCA - a powerful mathematical optimization algorithm - is integrated to accurately and quickly solve non-convex continuous constraints [2]. This combination promises to significantly optimize both physical quality and sample efficiency compared to pure machine learning methods.

1.2 Motivation (Why Japan & Why This Laboratory)

In the 1980s, Japan stood as an undeniable global superpower in semiconductor manufacturing, capturing approximately 50-60% of the world's market share. During this golden era, pioneering corporations such as NEC, Toshiba, and Mitsubishi Electric represented the pinnacle of technological prosperity and innovation in the global electronics industry. This profound historical legacy, combined with Japan's ongoing commitment to high-tech research, makes it the ideal environment for my academic pursuits in Electronic Design Automation. Furthermore, through careful investigation of your academic work, I have found that your laboratory has established robust strengths and impressive publications in Reinforcement Learning. This expertise aligns perfectly with my research interests and provides the exact guidance needed to successfully execute my proposed hybrid DRL-MCTS framework for chip macro placement.

2 Applicant's Background and Previous Research

During my academic career majoring in Computer Science at Hanoi University of Industry, I have built a solid foundation in Machine Learning and Deep Learning, with a specific focus on complex signal processing and optimization. My previous research primarily concentrated on Voice Processing, particularly Speaker Verification and voice biometrics. I have extensive experience designing, tuning, and optimizing advanced neural network architectures, such as ECAPA-TDNN and ResNet, and have actively

explored modern techniques including Kolmogorov-Arnold Networks (KAN). Moreover, I developed a hybrid TimeKAN-CNN approach for speaker verification and contributed significantly to the creation and refinement of the large-scale ViVoice34 dataset.

Designing these intricate deep learning architectures and processing high-dimensional datasets required rigorous mathematical optimization, algorithmic efficiency, and a deep understanding of complex network topologies. This background serves as a robust stepping stone for tackling the physical design of microchips. Furthermore, my transition into this domain is strongly motivated by your laboratory's pioneering contributions to reinforcement learning. Your comprehensive research on the intersection of deep learning, reinforcement learning, and world models highlights critical pathways for achieving sample-efficient learning in complex environments [3]. Additionally, your groundbreaking work on Distributional Pareto-Optimal Multi-Objective Reinforcement Learning provides a powerful framework for balancing conflicting objectives under uncertainty [4]. Combining my practical expertise in deep learning optimization with your lab's advanced RL methodologies [3, 4] will be instrumental in successfully executing my proposed hybrid DRL-MCTS framework for the multi-objective Chip Macro Placement problem.

3 Research Goals and Objectives

Research Goal

To propose a hybrid framework combining Deep Reinforcement Learning, Monte Carlo Tree Search (DRL - MCTS), and the DCA algorithm to optimize the quality of Chip Macro Placement (reducing HPWL and Congestion) while significantly improving learning efficiency and generalization capabilities.

Research Questions

1. How can the global decision-making mechanism of MCTS, guided by DRL, be effectively combined with the continuous optimization algorithm DCA in the macro placement problem?
2. How does the support of global tree search (MCTS) help reduce the number of required training samples for DRL and increase generalization capabilities on unseen chip designs?
3. How does the DCA algorithm help increase convergence speed and accuracy when handling non-linear geometric constraints (avoiding macro overlaps) compared to forcing the RL model to learn these constraints by itself?

4 Research Methodology

The proposed solution framework will be built based on the decomposition of the Chip Placement problem into two mutually supporting optimization layers:

- **Global Decision Layer (DRL + MCTS):** Utilizing Monte Carlo Tree Search combined with convolutional neural networks of DRL to search for the optimal placement sequence and spatial partitioning for macros. MCTS provides the model with "look-ahead search" capability, thereby enhancing generalization when facing netlists with entirely new topologies.
- **Geometric Optimization Layer (DCA):** After obtaining the rough partitioning and sequence from the upper layer, the exact coordinates of the macros will be determined by solving a non-convex continuous optimization problem. This problem (with the objective of optimizing HPWL, placement density, and completely eliminating overlaps between macros) will be effectively modeled and solved using the DCA [2].
- **Empirical Evaluation:** Utilizing public standard datasets such as ISPD, ICCAD, and Circuit Training. Key evaluation metrics will include HPWL, Congestion, Overlap ratio, required number of training samples, and inference time. (Note: Throughout the research, no sensitive personal data will be used).

5 Expected Outcomes and Career Path

- A complete hybrid algorithm framework capable of simultaneously solving the discrete combinatorial decision problem and the continuous geometric optimization problem for Chip Macro Placement.
- An empirical report demonstrating the superiority of this hybrid method over pure RL baselines in terms of both sample efficiency and generalization capability.
- Publishing research results in prestigious international conferences/journals on EDA or Artificial Intelligence (e.g., DAC, ICCAD, IEEE TCAD, or NeurIPS).
- A Master’s thesis serving as a solid foundation for further research at the Ph.D. level.
- **Future Career Path:** Upon completing my Master’s degree, my primary goal is to pursue a Ph.D. in Japan, aiming for a career as an R&D expert in semiconductor optimization and artificial intelligence. Beyond academics, I am deeply committed to immersing myself in Japan’s rich cultural heritage and advancing my current Japanese language proficiency to foster deeper, more authentic connections within the academic and local communities. By collaborating with Japanese researchers and bridging the technological sectors of our two nations, I strive to become a cultural and scientific ambassador. Through these continuous efforts, I hope to contribute meaningfully to the enduring diplomatic relationship between Vietnam and Japan, a strong and collaborative partnership that has flourished over the decades since its formal establishment in 1973.

6 Implementation Timeline

Timeline	Key Activities
Months 1–6 (Semester 1)	Literature review and completion of compulsory coursework; Building the simulation environment and testing existing open-source baselines.
Months 7–12 (Semester 2)	Designing the hybrid DRL-MCTS architecture integrated with the DCA algorithm; Preliminary training and experimental evaluation.
Months 13–18 (Semester 3)	Performing hyperparameter optimization; Conducting large-scale experiments on standard benchmarks; Writing scientific papers to submit to prestigious international conferences/journals.
Months 19–24 (Semester 4)	Finalizing the optimal model; Focusing on writing the Master’s thesis and preparing for the thesis defense.

7 References

References

- [1] Mirhoseini, A., et al. (2021). *A graph placement methodology for fast chip design*. Nature, 594(7862), 207-212.
- [2] Pham D.T., Le T.H. (2014). *The DC (Difference of Convex functions) Programming and DCA revisited with DC models of real world nonconvex optimization problems*. Annals of Operations Research, 225, 315–390.
- [3] Matsuo, Y., LeCun, Y., Sahani, M., Precup, D., Silver, D., Sugiyama, M., Uchibe, E., & Morimoto, J. (2022). *Deep learning, reinforcement learning, and world models*. Neural Networks, 152, 267-275.
- [4] Cai, X. Q., Zhang, P., Zhao, L., Bian, J., Sugiyama, M., & Llorens, A. J. (2023). *Distributional Pareto-Optimal Multi-Objective Reinforcement Learning*. Advances in Neural Information Processing Systems, 36.